

CALENDER

1. Day is the smallest unit of the calendar.
2. Odd days is the concept to solve calendar problems.

ODD DAYS:-

* The no. of days left over after converting the days into weeks.

* 200 days $\Rightarrow$ 28 weeks + 4 odd days $\left[\frac{200}{7} \right]$.

Leap & Non-leap years:-

* Leap year consists of 366 days $\Rightarrow$ 52 weeks + 2 odd days.

* Non leap year consists of 365 days $\Rightarrow$ 52 weeks and 1 odd day.

eg: 100th, 200th, 300th,

* Every century is not a leap year but

* Every 4th century year is a leap year.

eg: 100, 200, 300, 500, 600, 900, etc... $\rightarrow$ N.L year

eg: 400, 800, 1200, 1600, etc... $\rightarrow$ Leap years

* In 100 years there is 5 odd days

* 100 years $\Rightarrow$ 24 leap years + 76 N.L years

$$\Rightarrow 24 \times 2 + 76 \times 1$$

$$= 76 + 48$$

$$= 124 \text{ convert to weeks}$$

$$= 5 \text{ odd days}$$

$$\begin{array}{r} 17 \\ 7 \overline{)124} \\ \underline{7} \\ 54 \\ \underline{49} \\ 5 \end{array}$$

* No. of odd days should always be less than 7.

* 800 years $\rightarrow$ 16 odd days $\rightarrow$ 3 odd days

* 300 years $\rightarrow$ 15 odd days $\rightarrow$ 1 odd day

* 400 years $\rightarrow$ 20 odd days + 1 odd day

$$= 21 \text{ odd days}$$

$$= 0 \text{ odd days}$$

(X)

 $\therefore 100 \rightarrow 5 \text{ odd days}$
 $200 \rightarrow 3 \text{ odd days}$
 $300 \rightarrow 1 \text{ odd days}$
 $400 \rightarrow 0 \text{ odd days}$
 $500 \rightarrow 5 \text{ odd days}$
 $600 \rightarrow 3 \text{ odd days}$
 $\vdots$

3. If

0	1	2	3	4	5	6
Sun	Mon	Tue	Wed	Thu	Fri	Sat

1. How many odd days are there in 1956 years?

Ans: 1 odd day

1956

↓

$$1600 + 300 + 56 \leftarrow 14L = 14 \times 2 = 28$$

↓

↓

↓

$$42NL = 42 \times 1 = 42$$

0

+

1

↓

0

70

 $\div 7 = 0$

$$\Rightarrow 0 + 1 + 0 \Rightarrow 1$$

2. Jan 1 2008 was Monday. What day of the week falls on Jan 1 2010?

Ans: Thursday

Jan 1 2008 $\rightarrow$ Jan 1 2010

(20.0)

↓

2008

(1.03)

↓

2009

 $\Rightarrow 3 \Rightarrow \text{Monday} + 3 \Rightarrow \text{Thursday}$

3. on 6th March 2007 Monday falls. What was the day of the week on March 6th 2004?

Friday
Ans: ~~Monday~~

March 6 2007 → Monday

March

$$-3(2006-1, 2005-1, 2004-1)$$

March 6 2004 → Friday

4. on what day of the week 15th Aug 1947 falls?

Ans: Friday

$$\begin{array}{r} 1949 \\ \swarrow \quad \searrow \\ 1600 \quad 300 \quad 49 \end{array} \left\{ \begin{array}{l} 11L = 11 \times 2 \\ 35NL = 35 \times 1 \end{array} \right\} = 57 \Rightarrow 1 \text{ odd}$$

$$0 + 1 + 1 = 2 + \frac{3}{\downarrow}$$

$$\Rightarrow 2 + 3 = 5 \downarrow \text{Friday}$$

$$\begin{array}{r} 31 \\ 28 \\ 31 \\ 30 \\ 31 \\ 30 \\ 31 \\ 15 \end{array} \quad \begin{array}{r} 32 \\ 7 \overline{) 227} \\ \underline{3} \end{array}$$

5. on what day of the week 26th Jan 1950 falls?

Ans: Thursday

$$\begin{array}{r} 1949 \\ \swarrow \quad \searrow \\ 1600 \quad 300 \quad 49 \end{array} \left\{ \begin{array}{l} 12L = 24 \\ 37NL = 37 \end{array} \right. \quad \begin{array}{r} 8 \\ 7 \overline{) 61} \\ \underline{5} \end{array}$$

$$0 + 1 + 5$$

$$11 \text{ odd days} = 4 \text{ odd days} \Rightarrow \text{Thursday}$$

6. on what day of the week 11th Feb 2006 falls?

Ans: - Saturday

$$\begin{array}{r} 2005 \\ 1600 + 400 + 4 + 14 \\ 0 + 0 + 0 + 1 \Rightarrow 1 \end{array}$$

$$\text{Jan} + 11 \Rightarrow 31 + 11$$

J	31	1
F	28	2
M	31	3
A	30	4
M	31	5
J	30	6
J	31	7
A	31	8
S	30	9
O	31	10
N	30	11
D	31	12
	365	

7. The calendar year for 2007 will be same for the year a) 2014 b) 2016 c) 2018 d) 2019.

* If the No. of odd days between any two years is 0, then those 2 years will have same calendar.

* No. of odd days = 0 means

multiple of 7

Ans: 2018

2007 → 1

2008 → 2

2009 → 1

2010 → 1

2011 → 1

2012 → 2

2013 → 1

2014 → 1

2015 → 1

2016 → 2

2017 → 1

14 → 0

So, next will be 2018 has same calendar.

• (or) * Take last 2 digits of gn year and divide it with 4.

* If the remainder is,

1. 0 $\rightarrow$ Gn year + 28

2. 1 $\rightarrow$ Gn year + 6

3. 2 (or) 3 $\rightarrow$ Gn year + 11.

$$\Rightarrow 2007 + 11 \Rightarrow 2018$$

Given year

Ans

$\Downarrow$

$\Downarrow$

loop year $\rightarrow$

loop year.

N.L year $\rightarrow$

N.L year.

8. What is the maximum possible difference between any two leap years?

Ans: 8 yrs

1696 $\xrightarrow{+4}$ 1700 $\xrightarrow{+4}$ 1704
 $\downarrow$ $\downarrow$ $\downarrow$
 leap N.L leap year

9) Same calendar months in any year can be

a) Jan, Oct

b) Aug, Dec

c) Feb, Sep

d) April, July

$\Downarrow$

* not possible

because of

Feb

has 1, N.L year

Ans: April, July

Aug $\Rightarrow$ 3

Sep $\Rightarrow$ 2

Oct $\Rightarrow$ 3

Nov $\Rightarrow$ 2
 10

31 = 0 x

Apr $\Rightarrow$ 2

May $\Rightarrow$ 3

June $\Rightarrow$ 2

0 = 0 ✓

- 10) The last day of the century cannot be
 a) Monday b) Thursday c) Friday d) Sunday

Ans: Thursday

100 $\rightarrow$ 5 [Friday] possible

200 $\rightarrow$ 3 [Wednesday] possible

300 $\rightarrow$ 1 [Monday] possible

400 $\rightarrow$ 0 [Sunday] possible.

Tuesday

Thursday

Saturday

These days

cannot be last day of
the century.

11. If today is Wednesday, what would be the day of week after 192 days?

Ans: Saturday

Today $\rightarrow$ Wednesday

+ 192 $\rightarrow$ Wednesday + 3 (OD) $\rightarrow$ Saturday.

12. If 266th day of a leap year is Friday. What would be the last day of the same leap year?

Ans: Sunday

266th $\rightarrow$ Friday

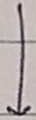
last day $\rightarrow$ Friday + 2 (OD's)

$\rightarrow$ Sunday.

13. If 13th Jun 2012 falls on Tuesday, what is the day on 13th Jun 2004.

Ans: Saturday.

13th June 2012



8 years $\Rightarrow$ 2 leap years + 6 N-leap years

13th June 2004 $= 2 \times 2 + 6 \times 1 = 10 \Rightarrow 3 \text{ (OD)}$

$\Rightarrow \text{Tues} - 3 \Rightarrow \text{Saturday}$

14. If 26th May 2021 falls on Saturday then what is the day on 26th May 2030.

Ans: Wednesday.

26th May 2021



9 years $\Rightarrow 2 \times 2 + 7 \times 1 = 11 = 4 \text{ OD}$

26th May 2030

$\Rightarrow \text{Saturday} + 4 \Rightarrow \text{Wednesday}$



15. How many days does 'k' weeks and 'k' days will have in total.

a) $6k$ days b) $7k$ days c) $8k$ days d) $10k$ days

Ans: $8k$ days

'k' week + 'k' days

$\Downarrow$
 $7k \text{ days} + k \text{ days} = 8k \text{ days}$

16. on what ~~date~~^{tes} of April 2001 did wednesday fall?

Ans: 4, 11, 18, 25

17. on what dates of Dec 2025 did Monday fall?

Ans: 1, 8, 15, 22, 29

16.) 1 Apr 2001

$$\begin{array}{r} 2000 \\ \downarrow \\ 0 \end{array} \quad \begin{array}{r} 31+28+31+1 \\ \downarrow \\ 91 \Rightarrow 0 \end{array}$$

1 April 2001 $\rightarrow$ Sunday

2 April $\rightarrow$ Mon

3 April $\rightarrow$ Tues

4 April $\rightarrow$ Wed

+7 11

+7 18

+7 25

17.) 1 Dec 2025

$$\begin{array}{r} 2024 \\ \downarrow \\ 0 \end{array} \quad \begin{array}{r} 31+28+31+30 \\ +31+30+31+31 \\ +30+31+30 \end{array}$$

2024

$$\begin{array}{r} 2000 \\ \downarrow \\ 0 \end{array} \quad \begin{array}{r} 24 \\ \downarrow \\ 2 \end{array}$$

$\Rightarrow 2 + 6 = 8 = 1 \text{ OD}$

Dec 1 2025

365

30

335

$\Rightarrow 6 \text{ (OD)}$

$2+6=8=1 \text{ OD}$

1 Dec 2025 - M

(+7) 8 Dec

(+7) 15 Dec

(+7) 22 Dec

(+7) 29 Dec